Forest Guardian

Satellite fire risk prediction — complete project details

Prepared 2026-08-19 · California pilot region

1. Product Overview

Forest Guardian is a website that uses an AI model to predict and display forest fire risk in real time, combining a model trained on historical fire data with live satellite and weather data. Built for government use, open to the public with no login required, English-language interface. The live implementation is branded **Canopy** on the map screen itself.

Core features (MVP)

- Real-time fire prediction map — organic risk zones per named forest region, not abstract circles/blur
- AI prediction model — trained on historical fire detections + weather, scoring live conditions
- Live active-fire detections from satellite, clustered into readable events
- Day-ahead forecast scrubber (today, +1, +2, +3 days)
- Recent fire-history section, sourced from real historical detections

2. Design System

Warm, organic, credible visual language centered on the Forest Guardian emblem — a flame-and-leaf badge with an orbiting satellite. Explicit principle: government-grade credibility without cold "command-center" aesthetics.

Ember Orange	Burnt Umber	Golden Amber	Cream Glow
#e8622c	#a8481f	#f0a34a	#ffedd2

- **Typography:** Righteous (display/wordmark), Quicksand or Nunito (body/UI)
- **Shape language:** rounded corners, organic hand-drawn zone shapes on the map (explicitly not circles)
- **Motion:** soft/organic easing echoing the logo's flicker/breathe/orbit animation, used sparingly
- **Constraint:** no purple anywhere; functional/risk colors stay in the warm amber→orange→red-orange family

3. Data Sources

Source	Used for	Status
NASA FIRMS (VIIRS NRT + SP archive)	Live active-fire detections; historical fire-occurrence labels (model v1)	**Live**
Open-Meteo (forecast + historical archive)	Live/forecast weather per zone; historical weather for model v1 training	**Live**
OpenStreetMap / Overpass API	California & Azerbaijan boundary polygons; initial forest/power-line/road asset survey	**Live (boundary use)**
MODIS MCD64A1 (burned area, via Earth Engine)	Cleaner fire-occurrence ground truth for model v2 (vs. FIRMS point detections)	**In progress**
GRIDMET (via Earth Engine)	Weather features for model v2, incl. 90-day trailing precipitation (drought signal)	**In progress**
ESA WorldCover v200 (via Earth Engine)	Forest-only land-cover masking of the model v2 grid	**Done (grid built)**
SRTM (via Earth Engine)	Terrain features: elevation, slope, aspect	**Done (grid built)**
Sentinel-2 / Sentinel-1	NDVI/NDMI vegetation dryness, SAR	**Tested, not yet in**

(Copernicus, via Earth Engine)	moisture — tested and confirmed working	**training set**

4. Modeling

Model v1 — live on the current site

- Logistic regression, trained client-side-inferable (single dot-product + sigmoid, no ML runtime in the browser)
- Labels: any VIIRS fire detection in a grid cell on a given day (California, 12×12 regional grid, 66 land cells)
- Features: daily temp max/min, max wind speed, precipitation sum, mean humidity (Open-Meteo)
- Training set: ~31,500 cell-day rows, ~15 months
- Result: test AUC 0.66 — real but modest signal, coefficients physically sensible

Model v2 — in progress

> Rebuilt after identifying that FIRMS-detection labels conflate real vegetation fires with gas flares, industrial heat, and desert/urban thermal anomalies. v2 uses MODIS burned-area (confirmed vegetation burn) instead, restricted to a forest-only grid, with terrain and historical fire frequency added as features. Serving moves from client-side logistic regression to a real backend running Random Forest / XGBoost.

- Grid: 3 km cells, forest-masked via ESA WorldCover (≥30% tree cover) — **24,383 real forest cells** across California
- Time range: 22 months (Aug 2024 – May 2026, limited by MCD64A1's ~3.5-month processing latency), monthly timestep
- Labels: MODIS MCD64A1 burned-area, any burned pixel in-cell that month
- Features: GRIDMET monthly weather (max/min temp, wind, min humidity, precipitation) + 90-day trailing precipitation, elevation, slope, aspect, forest fraction, prior fire count (rolling historical frequency)

- Models: Random Forest and XGBoost, class-weighted for the rare-event imbalance
- Evaluation: precision, recall, F1, AUC-PR (not just accuracy/AUC-ROC, which are misleading for imbalanced classes)
- Status at time of writing: grid built, feature export running (~22,000+ cell-months), training not yet run

5. Application Architecture

Frontend

- Single self-contained page (`app/index.html`) — no React/build-step dependency for the page itself
- Leaflet for the map; organic hand-generated zone polygons per named region (not circles/heatmap blur)
- Three.js hero globe (decorative), D3/topojson for the globe's world map texture
- Live data wiring: trained model run client-side against live Open-Meteo weather per zone; live FIRMS detections clustered into readable fire events with real wind direction; fire-history section from a precomputed real-data summary

Backend

- `firms-proxy.js` + `server.js` — keeps the FIRMS API key server-side only, proxies `/api/firms` in both dev (Vite middleware) and production (standalone Node server)
- Planned: small Python service serving Random Forest / XGBoost predictions, re-scoring all forest cells on a **daily** cadence (matching GRIDMET's own update frequency) and serving a cached snapshot — not per-request Earth Engine calls, to keep site latency near-instant

Run commands

Command	Purpose
`npm run dev`	Local dev server (default port 5183 in this session), proxies

	`/api/firms`
`npm run build && npm start`	Production build + standalone server on port 4173

6. Current Status

Component	Status
Data fetch pipeline (FIRMS, Open-Meteo, OSM boundaries)	**Live**
Canopy map screen (frontend)	**Live**
Model v1 (logistic regression, client-side)	**Live, currently serving the site**
FIRMS key proxy (dev + prod)	**Live**
Earth Engine access (Sentinel, MODIS, GRIDMET, WorldCover, SRTM)	**Authenticated & confirmed working**
Model v2 forest grid (24,383 cells)	**Built**
Model v2 feature export (MODIS + GRIDMET, 22 months)	**Running**
Model v2 training (RF/XGBoost)	**Not yet run**
New backend to serve model v2	**Not yet built**
Region expansion beyond California (e.g. Europe)	**Deferred until v2 is validated**

7. Known Limitations

- Model v1's "fire day" label is whole-region daily aggregate, not spatially fine-grained — this is the primary motivation for model v2

- FIRMS confidence codes (VIIRS: low/nominal/high) are mapped to approximate percentages (35/70/92%), not a true calibrated probability
- Point fire detections don't give containment status, burned hectares, or a name — the app derives an honest "nearest monitored region" label and total radiative power (MW) instead of fabricating those fields
- Model v2's fire-occurrence prediction still has an irreducible random component — most real ignitions are human-caused or lightning-triggered, not fully predictable from weather/terrain/vegetation alone
- Vegetation (Sentinel-2/1) features tested and confirmed working via Earth Engine, but not yet incorporated into the model v2 training set

8. Repository Structure

Path	Contents
`app/`	The live site: `index.html` (Canopy), `server.js`, `firms-proxy.js`, `vite.config.js`, `public/data/` (model weights, fire history)
`model/`	Model v1 pipeline: historical data fetch, spatial training-set builder, trainer, trained weights
`model/gee_pipeline/`	Model v2 pipeline: forest grid builder, MODIS/GRIDMET feature exporter, RF/XGBoost trainer
`data/`	Original raw data pulls (FIRMS, Open-Meteo, OSM assets) from the initial project setup
`.env`	FIRMS_MAP_KEY (gitignored, never committed)

Generated from the project's working state as of this session. Model v2 accuracy numbers were not yet available at time of writing — training had not completed.